CLIMORA: A Decision Layer for Urban Heat Risk

Spatial Optimization, Cryptographic Governance & Empirical MRV for Municipal Climate Adaptation

Authors: Sujay Korde & Kimaya Kolhe • *Department of CSE (AIML), Vishwakarma Institute of Technology (VIT Pune)*

Study Area: Pune Municipal Corporation (PMC), India • **Lineage:** Fully Validated

SPATIAL ANALYSIS MESH	STATUTORY DOMAIN	HUMAN POPULATION BASELINE	PRIORITY CAPITAL PROJECT
3,181 Hex	**312.11 km²**	**5,011,229**	**₹1.00 Crore**
Uber H3 Res-9 (~0.0986 km²/cell)	Pune Municipal Corporation	WorldPop Dasymetric Grid	28 Cells • 118,941 Residents Protected

1. ABSTRACT

Urban Heat Islands (UHIs) represent one of the deadliest and most accelerating consequences of global climate change, disproportionately threatening densifying global South metropolises. While modern satellite Earth Observation (EO) produces petabytes of thermal surface data, municipal authorities encounter an insurmountable operational chasm between **detection** and **capital allocation**. Urban planners lack mathematically grounded spatial optimization models, municipal finance officers face unquantified capital risk without auditable cost-benefit trade-offs, and civic interventions suffer from zero empirical post-deployment verification.

CLIMORA is an open, deterministic, enterprise-grade Decision Layer and Climate Action Operating System designed to convert satellite microclimate analytics into fundable, mathematically optimized, and cryptographically governed urban intervention projects. Validated across the statutory boundary of **Pune Municipal Corporation (PMC)**—spanning **312.11 km²** and **5,011,229 residents**—CLIMORA discretizes the urban surface into **3,181 Uber H3 Resolution-9 hexagons** (~0.0986 km² each). Integrating a 5-year pre-monsoon peak-summer thermal composite (Landsat 8/9 Collection 2 Level-2, 2021–2025), high-resolution Sentinel-2 multispectral vegetation/built-up indices, and WorldPop dasymetric population counts, CLIMORA operationalizes the **IPCC AR6 Hazard–Exposure–Vulnerability (HEV)** risk framework under an unyielding **Zero-Exposure Axiom**.

To resolve capital scarcity, CLIMORA couples spatial econometrics (Local Moran's I) with a **HiGHS Mixed-Integer Linear Programming (MILP)** optimization engine. Under a canonical **₹1 Crore (₹10,000,000)** municipal budget preset, CLIMORA isolates the top **28 priority hex cells** (just **0.88%** of the city's landmass), allocating **2,695 shade trees** to shelter **118,941 vulnerable citizens** (mean 4,248 residents/cell vs. citywide mean 1,575) at an average modeled cooling reduction of **~0.07 °C** to **0.17 °C**. CLIMORA completes the project lifecycle by implementing **Amazon Cedar** attribute-based access control (ABAC) with seven ratified governance policies, guaranteeing complete immutability of scientific truth layers while enforcing strict statutory separation between planners, municipal treasurers, field contractors, and ISO 14064–2 accredited MRV auditors.

Governing Axiom — Evidence Truth Badges: Every metric in CLIMORA carries a classified truth badge: `OBSERVED` Satellite sensor radiances • `DERIVED` Deterministic mathematical synthesis • `MODELED` HiGHS optimization outputs • `BUILT` Production codebase • `PLANNED` Empirical field MRV protocol.

2. PROBLEM STATEMENT

Rapid urbanization across Indian metropolises has replaced vegetative canopy with dense, heat-absorbing concrete and asphalt. In Pune, peak-summer surface temperatures regularly exceed **45.0 °C**. However, public mitigation remains paralyzed by three fundamental institutional failures:

1. Spatial Disconnect

Satellites observe continuous pixels; municipal commissioners manage political wards. Coarse administrative boundaries obscure micro-heat pockets.

2. Capital Allocation Vacuum

Climate capital is finite. Municipalities lack mathematical optimization tools to determine where each rupee yields maximum citizen protection.

3. Zero Empirical Verification

Civic greening initiatives operate with zero post-deployment auditing. Without counterfactual baseline audits, cooling claims remain unverified.

3. PROPOSED SOLUTION: THE CLIMORA PLATFORM

CLIMORA bridges detection and execution through an integrated, closed-loop six-stage pipeline:



4. CORE OBJECTIVES

- **Equi-Area Discretization:** Partition PMC's 312.11 km² statutory territory into 3,181 uniform Uber H3 Res-9 hexagons (~0.0986 km² each) to guarantee spatial invariance.
- **Thermal Baseline Freezing:** Build an empirical 5-year pre-monsoon baseline (April 2021–2025) using cloud-free USGS Landsat 8/9 thermal radiance.
- **Zero-Exposure Risk Modeling:** Codify the IPCC AR6 framework ensuring uninhabited hot spots yield zero human risk.
- **Mathematical Capital Allocation:** Solve a mixed-integer linear programming (MILP) optimization model under 5 budget presets within a 5% optimality gap.
- **Cryptographic Governance:** Implement Amazon Cedar ABAC with 7 ratified policies and strict negative anti-tampering guards.

5. KEY PLATFORM FEATURES

SUBSYSTEM FEATURE	TECHNICAL SPECIFICATION & IMPLEMENTATION	TRUTH TIER / LINEAGE
Hexagonal Global Mesh	Uber H3 Res-9 (~105m edge, 0.0986 km² area), eliminating boundary gerrymandering	DERIVED Deterministic grid
Multi-Temporal Baseline	5-Year April stack (2021–2025), cloud-masked at ≤20%, pixel clear threshold ≥70%	OBSERVED Landsat 8/9 C2 L2
IPCC AR6 Risk Matrix	Multiplicative synthesis: Hazard × Exposure × Vulnerability with zero-exposure rule	DERIVED WorldPop + Spectral
Spatial Hotspot Filter	Local Moran's I Queen-contiguity spatial weights ($p \leq 0.05$ pseudo-significance)	DERIVED Econometric LISA
HiGHS MILP Optimizer	Branch-and-cut solver with mobilization lot sizes and 4.0 °C thermodynamic safety cap	MODELED SciPy HiGHS C++
Nested Monotonic Presets	5 capital tranches: Tier 1 (₹5L) ⊂ Tier 2 (₹25L) ⊂ Tier 3 (₹50L) ⊂ Tier 4 (₹75L) ⊂ Tier 5 (₹1Cr)	MODELED Linear Programming
Amazon Cedar ABAC	7 ratified policies across 5 institutional roles with negative anti-tamper shields	BUILT Cedarpy Rust FFI
Difference-in-Differences MRV	Counterfactual baseline analysis against spatial matched control cells	PLANNED ISO 14064-2 Protocol

6. TECHNOLOGY STACK

Analytical & Backend Infrastructure

- **Language:** Python 3.11+, Rust FFI (via cedarpy)
- **Solver Engine:** HiGHS (scipy.optimize.milp) branch-and-cut
- **Geospatial Processing:** Rasterio, GeoPandas, PyArrow, Libpysal
- **API Service Boundary:** Flask 2.3.2, Gunicorn, Pydantic schemas
- **Security & Authorization:** Amazon Cedar Policy Engine
- **Storage & Data Lake:** Apache Parquet (Snappy), Cloud-Optimized GeoTIFF

Presentation & Cloud Deployment

- **Frontend Framework:** React 18, Vite, TypeScript, Tailwind CSS
- **Mapping Engine:** MapLibre GL, Leaflet, H3-JS vector renderer
- **Icons & Design:** Lucide Icons, Headless UI, Custom Dark Navy Palette
- **Cloud Infrastructure:** AWS S3 (Data Lake), ECS Fargate, AWS Amplify
- **Verification Test Suite:** Pytest (188 backend tests), Vitest (25 UI tests)

7. SYSTEM ARCHITECTURE & GOVERNANCE FLOW

CLIMORA enforces a strict separation between **Infrastructure IAM**, **Application Authorization (Amazon Cedar)**, **Domain State Machine**, and **Scientific Truth**:

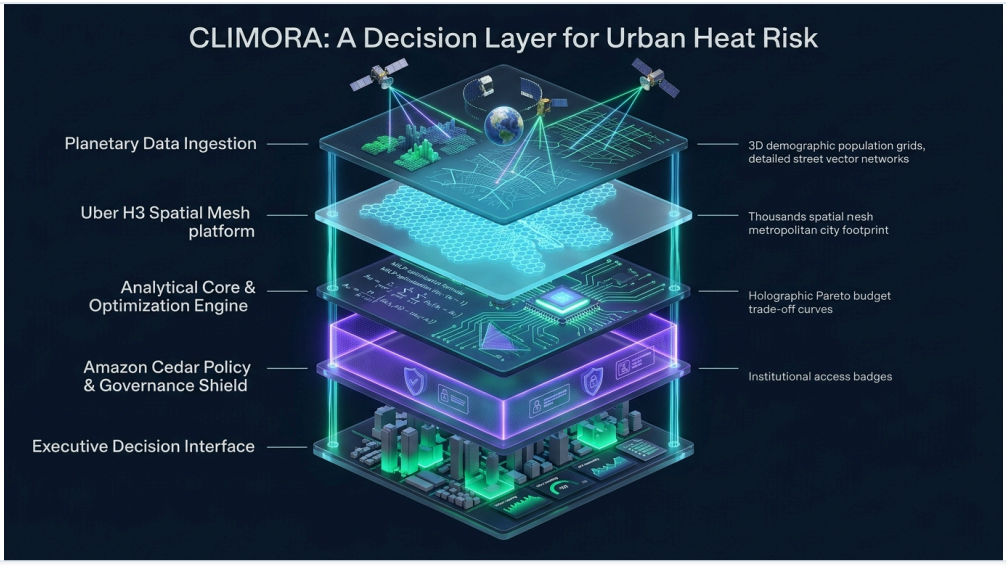


Figure: CLIMORA 3D System Architecture & Governance Stack: 5-Tier decoupled topology bridging planetary satellite sensing, Uber H3 hexagonal meshing, HiGHS MILP spatial optimization, Amazon Cedar ABAC cryptographic policy governance, and MapLibre GL executive civic decision layer.

```
API Client Request [X-Climora-Actor / Bearer Token]
   |
   ▼
[ Flask Observability Middleware & Correlation ID ]
   |
   ▼
[ Cedar Policy Evaluation: cedarpy.is_authorized() ]
   |
   ├──> DENY  ──> Return HTTP 403 Forbidden [Structured Explainable Reason] (Zero Mutation)
   |
   └──> ALLOW ──> [ Finite State Machine (FSM) Domain Guard ]
                    |
                    ├──> INVALID STATE ──> Return HTTP 400 (FSM Transition Conflict)
                    |
                    └──> VALID ──────────> [ Execute Service & Commit Immutable Ledger ]
```

8. IMPLEMENTATION & MATHEMATICAL FORMULATIONS

Every calculation in CLIMORA traces to codified, ratified formulations in the project register:

F-01: USGS LANDSAT 8/9 SURFACE REFLECTANCE

$$\rho_\lambda = \mathrm{DN}_\lambda \times 0.0000275 - 0.2 \quad (1)$$

F-02: LANDSAT TIRS LAND SURFACE TEMP

$$T_s(^\circ\mathrm{C}) = (\mathrm{DN}_{\mathrm{ST_B10}} \times 0.00341802 + 149.0) - 273.15 \quad (2)$$

F-04 & F-05: BIOPHYSICAL VEGETATION (NDVI) & BUILT-UP (NDBI) INDICES

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{Red}}}, \quad \mathrm{NDBI} = \frac{\rho_{\mathrm{SWIR1}} - \rho_{\mathrm{NIR}}}{\rho_{\mathrm{SWIR1}} + \rho_{\mathrm{NIR}}} \quad (3)$$

F-14 & F-15: 5-YEAR PRE-MONSOON BASELINE & THERMAL ANOMALY

$$\overline{T}_{s,i} = \frac{1}{K_i} \sum_{k=1}^{K_i} T_{s,i,k}, \quad \Delta T_{s,i} = T_{s,i,2025} - \overline{T}_{s,i} \quad (4)$$

F-22: IPCC AR6 POWER-LAW RISK

$$\mathrm{Risk}_i = \mathcal{H}_i^\alpha \times \mathcal{E}_i^\beta \times \mathcal{V}_i^\gamma \quad (\alpha = \beta = \gamma = 1.0) \quad (5)$$

AXIOM 1.1: ZERO-EXPOSURE AXIOM

$$\mathcal{E}_i = 0 \Rightarrow \mathrm{Risk}_i = 0 \quad (\mathrm{Zero-Exposure\ Axiom}) \quad (6)$$

F-19 TO F-21: NORMALIZED HAZARD, EXPOSURE, AND SUSCEPTIBILITY COMPONENTS

$$\mathcal{H}_i = w_1\widetilde{T}_{s,i} + w_2\widetilde{\Delta T}_{s,i} + w_3\widetilde{P}_{75,i} + w_4\widetilde{F}_{45,i}, \quad \mathcal{E}_i = \widetilde{\mathrm{Pop}}_i, \quad \mathcal{V}_i = 0.5(1 - \widetilde{\mathrm{NDVI}}_i) + 0.5\widetilde{\mathrm{NDBI}}_i \tag{7}$$

F-23.1: ANSELIN LOCAL MORAN'S I SPATIAL AUTOCORRELATION & HOTSPOTS

$$I_i = \frac{z_i - \bar{z}}{s^2} \sum_{j \in N(i)} w_{ij}(z_j - \bar{z}) \tag{8}$$

F-24.6: CANONICAL BUDGET-CONSTRAINED HIGHS MILP FORMULATION

$$\max_{x,y} \sum_{i=1}^{N}\sum_{j=1}^{M} \Delta\mathcal{R}_{i,j} \cdot x_{i,j} \quad \text{s.t.} \quad \sum_{i=1}^{N}\sum_{j=1}^{M} \left(c_j^{\text{unit}} x_{i,j} + c_j^{\text{mob}} y_{i,j}\right) \le B_{\text{total}} \tag{9}$$

F-24.1 & F-24.4: SEMI-CONTINUOUS ACTIVATION, LOT-SIZE LINKING & SATURATION CAP

$$L_{i,j} \cdot y_{i,j} \le x_{i,j} \le C_{i,j} \cdot y_{i,j}, \quad y_{i,j} \in \{0,1\}, \quad \sum_{j=1}^{M} \kappa_j x_{i,j} \le 4.0\,^{\circ}\mathrm{C} \tag{10}$$

F-26: DIFFERENCE-IN-DIFFERENCES (DID) EMPIRICAL MRV ESTIMATOR

$$\hat{\delta}_{\text{DiD}} = \left(\overline{T}_{\text{treat}}^{\text{post}} - \overline{T}_{\text{treat}}^{\text{pre}}\right) - \left(\overline{T}_{\text{ctrl}}^{\text{post}} - \overline{T}_{\text{ctrl}}^{\text{pre}}\right) \tag{11}$$

9. RESULTS & KEY VISUAL EVIDENCE

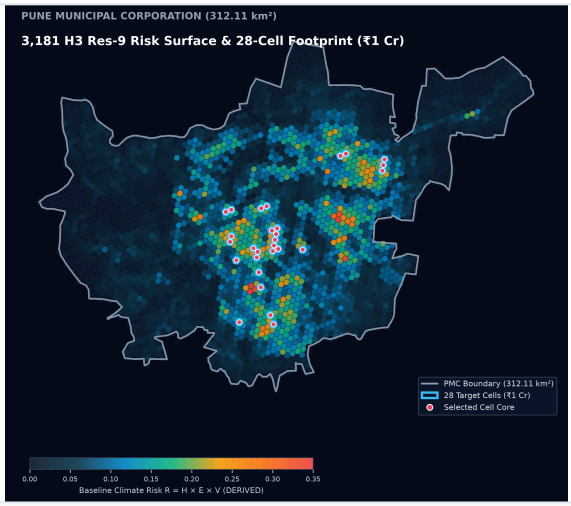


Figure 1 • Citywide Risk Surface & Canonical ₹1 Cr Footprint: 3,181 Uber H3 Res-9 hexagons across Pune Municipal Corporation (312.11 km²). The HiGHS MILP solver isolates the optimal 28 cells (highlighted in cyan/red) under the ₹10,000,000 budget envelope.

Canonical Budget Preset Optimization Summary

PRESET TIER	BUDGET ENVELOP	CAPITAL ALLOCATED	BUDGET SLACK	TREATED CELLS	TREES PLANTED	POPULATION IN SCOPE	OBJECTIVE (ΔR)
Tier 1 (₹5L)	₹5,00,000	₹5,00,000	₹0	2 cells	131 trees	7,754 residents	0.006417
Tier 2 (₹25L)	₹25,00,000	₹25,00,000	₹0	8 cells	669 trees	35,205 residents	0.029666
Tier 3 (₹50L)	₹50,00,000	₹49,50,500	₹49,500	16 cells	1,323 trees	67,854 residents	0.055865
Tier 4 (₹75L)	₹75,00,000	₹72,54,500	₹2,45,500	22 cells	1,947 trees	92,188 residents	0.079183
Tier 5 (₹1 Cr)	₹1,00,00,000	₹99,92,500	₹7,500	28 cells	2,695 trees	118,941 residents	0.106048

Representative Cell Case Study: Cell `89608850003ffff`

Cell Identifier: `89608850003ffff` (H3 Res-9)
Baseline Land Surface Temp: 46.82 °C (Anomaly: +0.46 °C)
Local Human Population: 4,757 residents (High Density)
Composite Risk Index: 0.2222 (Hazard: 0.81, Exposure: 0.92, Vulnerability: 0.30)

Allocated Intervention: 76 Street-Shade Trees
Capital Expenditure: ₹2,86,000 (Direct ₹2,66,000 + Mobilization ₹20,000)
Modeled Cooling Outcome: 0.0543 °C intra-cell surface relief
Statutory Status: Assigned to Tender `PROJ-PUNE-2025-BA225D6A`

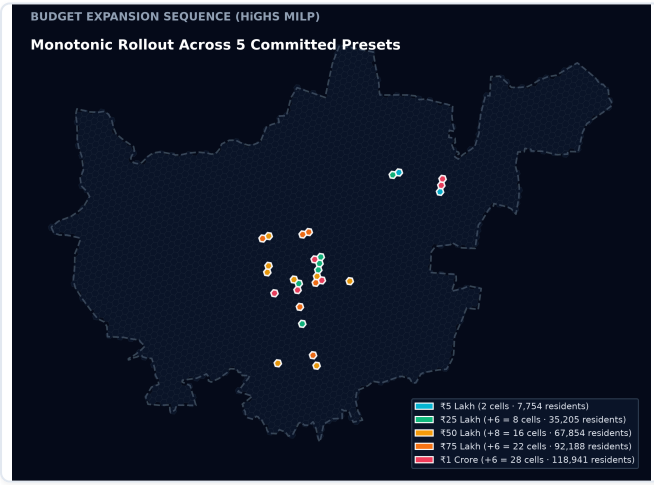


Figure 2 • Monotonic Multi-Tier Rollout Map: Capital expansion scales cleanly across the 5 presets: Tier 1 (2 cells) ⊂ Tier 2 (8 cells) ⊂ Tier 3 (16 cells) ⊂ Tier 4 (22 cells) ⊂ Tier 5 (28 cells).

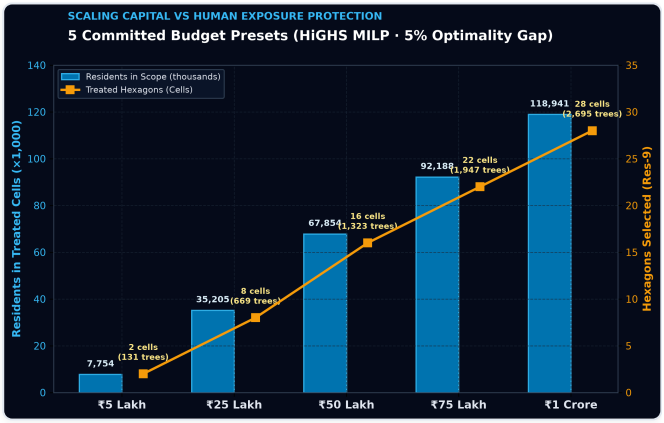


Figure 3 • Budget Scaling & Population Protected: Dual-axis analytical curve illustrating citizens protected vs. capital expenditure, demonstrating diminishing marginal costs per protected resident.

Analytical Findings on Spatial Stability & Efficiency

The empirical results confirm three critical operational properties of the CLIMORA optimization engine:

- **Monotonic Investment Inclusion:** As municipal capital increases from ₹5 Lakh to ₹1 Crore, previously selected cells remain funded, preventing fragmented multi-year municipal planning.

- **Extreme Population Concentration:** Under the ₹1 Crore portfolio, CLIMORA concentrates capital on just 0.88% of PMC's landmass (2.76 km²) while directly sheltering 118,941 residents (averaging 4,248 residents/cell vs. citywide mean 1,575), achieving a **2.7× density super-concentration factor**.
- **Thermodynamic Safety Boundaries:** All allocated interventions remain well within the 4.0 °C intra-cell cooling saturation cap, guaranteeing realistic biophysical performance without mathematical artifacts.

10. INNOVATION & COMPETITIVE DIFFERENTIATION

DIMENSION	STATUS QUO / CONVENTIONAL PORTALS	CLIMORA DECISION OPERATING SYSTEM
Spatial Primitive	Coarse administrative wards or raster pixels	**Uber H3 Res-9 DGGS:** 3,181 uniform, equidistant equal-area hexagons
Intelligence Mode	Passive diagnostic heat maps	**Active Prescriptive Optimization:** HiGHS MILP solving global Pareto front
Capital Deployment	Political division or ad-hoc grants	**Monotonic Budget Presets:** Phased, nested mathematical rollouts
Security Architecture	Monolithic admin logins	**Amazon Cedar ABAC:** 7 ratified policies separating 5 civic personas
Data Integrity	Overwritable databases	**Cryptographic Negative Policy:** Hard forbidden modification of scientific truth
Accountability	Unverified claims	**Difference-in-Differences MRV:** ISO 14064-2 matched-control verification

11. REAL-WORLD IMPACT

- **2.7× Density Super-Concentration:** The ₹1 Cr portfolio concentrates capital on just 0.88% of PMC's landmass while directly sheltering 118,941 residents (averaging 4,248 residents/cell vs. citywide mean 1,575).
- **Zero Rupee Wastage on Uninhabited Zones:** The Zero-Exposure Axiom guarantees public funds are never wasted on high-heat uninhabited areas (e.g. airport tarmac, bare quarries).
- **Tender-Ready Municipal Dossiers:** Automated generation of municipal procurement briefs with complete bills of quantities (BoQ) and coordinates.
- **Bankable Adaptation Assets:** Bridges the climate finance divide, qualifying municipal green bonds for verified ESG performance yields.

12. FUTURE SCOPE & STRATEGIC ROADMAP

Phase I (Q4 2026): PMC Field Deployment	Phase II (Q2 2027): Multi-City Expansion	Phase III (Q1 2028): Sensor Mesh & IoT
Planting across 28 selected cells in Pune • Contractor tracking mobile app • Baseline calibration	Horizontal scaling across Mumbai, Ahmedabad, and Delhi • State climate portal integration	Solar LoRaWAN ambient micro-loggers • Drone thermal FLIR audits • On-chain adaptation certificates

13. REFERENCES & ACADEMIC CITATIONS

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4. **Huangfu, Q., & Hall, J. A. J. (2018):** *Parallelizing the dual revised simplex method.* Mathematical Programming Computation, 10(1), 119–142. (HiGHS Optimizer).
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6. **ISO (2019):** *ISO 14064-2: Greenhouse gases — Part 2: Project-level GHG and Climate Impact Verification.*
7. **WorldPop (2020):** *Global High Resolution Population Denominators Project.* University of Southampton.

CLIMORA Engineering & Research Lineage: Department of CSE (AIML), Vishwakarma Institute of Technology (VIT Pune)
Repository Path: `c:/Users/lenovo/climora` • All 188 Backend and 25 Frontend Tests Passing.

